A Rule-Constrained Large Language Model Pipeline for Rule-Consistent Explanation of Vietnamese Complete Blood Count Results

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Abstract. Vietnamese patients increasingly ask general-purpose language models to explain laboratory reports, yet an unconstrained model can misstate results or falsely reassure. This study evaluates a rule-constrained pipeline in which deterministic rules assign every clinical label before invoking a language model; the model only paraphrases those labels, and a post-generation validator checks the output. On 150 synthetic complete blood count panels with known labels, an unconstrained Qwen2.5-3B model misstated at least one analyte direction in 91.3% of panels (95% CI 85.7–94.9) and falsely reassured in 32.0% (25.1–39.8). Prompt constraints reduced these rates to 12.7% and 8.0%; adding validation left no output that violated the implemented checks, with 64.0% of panels served by model text and 36.0% by fixed templates. A deterministic template-only configuration reaches the same pass rate with no model at all, so the measure reports rule consistency under the implemented checks rather than clinical safety, and what the model adds is the 64.0% of panels that keep generated prose. The system runs on a CPU without a GPU. All results come from a single 3-billion-parameter model served locally, so they bound what a small on-premises model does, not what the larger hosted services patients typically use would do. Readability and patient comprehension were not measured. The contribution is an auditable architecture and a separately reported false-reassurance metric, not evidence of clinical benefit.

Keywords: health informatics, health literacy, large language models, laboratory results, output validation, patient safety.

1 Introduction

A patient leaving a Vietnamese laboratory receives a panel of values annotated with terms such as PDW, MCHC, and Monocyte, in units such as G/L, fL, and pg. The

reference interval shows whether a value is out of range, but not what the analyte measures or whether the deviation matters. Health literacy remains limited in Vietnam, including among digitally connected adults [1–4], and patients who receive results without explanation often search online [5]. Increasingly, that search means pasting the report into a general-purpose language model.

Such models can fail in two ways. They can misstate facts: asked in Vietnamese what white blood cells are, the 3B model used here answered that they are lymphocytes and, unprompted, mentioned cancer. More consequentially, they can describe an abnormal panel as normal. This false reassurance is fluent and plausible, and it removes the reason to seek care, so we treat it as a distinct error class.

Three questions guide the study: (RQ1) How often does unconstrained generation misstate results or falsely reassure on panels with known labels? (RQ2) How much do rule-assigned labels and prompt constraints reduce these errors? (RQ3) Can a post-generation validator eliminate outputs that violate the implemented checks, and what does that cost in fallback rate and latency? The study is an internal evaluation on synthetic data, not a clinical trial, and it does not measure whether patients understand the output.

2 Related Work and Contributions

Studies of language models for laboratory results mainly evaluate answer quality. He et al. [6] compared model and peer answers to patient questions, while Munoz-Zuluaga et al. [7] and Cadamuro et al. [8] reported incorrect or superficial answers and advised against diagnostic use. Similar assessments cover radiology reports [9] and scientific articles for patients [10]. Table 1 positions the present design against these studies; it is a selective comparison, not a systematic review.

Table 1. Comparison of related studies on language models explaining medical results.

Study	**Data and design**	**Reported evidence**	**Comparison with this study**
He et al. (2024) [6]	53 patient questions from a health forum; LLMs vs. peer answers.	Relevance, accuracy, helpfulness, harm; GPT-4 best.	Here: rule-assigned labels and validated output.
Munoz-Zuluaga et al. (2023) [7]	30 physician-written interpretation questions; GPT-4.	46.7% correct, 23.3% partially correct, 30% incorrect.	Here: patient-facing; false reassurance measured.
Cadamuro et al. (2023) [8]	EFLM working-group assessment of ChatGPT on test results.	Superficial or incorrect answers; not for diagnosis.	Here: clinical judgement removed from the model.
Bozer and Pekçevik (2025) [9]	Radiology reports explained by several LLMs; expert rating.	Readability, understandability, communication.	Here: consistency with assigned labels.
This study	150 synthetic CBC panels; 3 configurations; Qwen2.5-3B on CPU.	Six error classes, fallback rate, latency.	Auditable pipeline; no clinical validation.

Output control is a second line of work. Grammar-constrained decoding restricts generation to strings valid under a grammar [11], guided generation makes schema-constrained output efficient [12], and knowledge-constrained decoding prunes hallucinated tokens [13]. Structural validity does not ensure semantic correctness, so our validator checks content against labels computed independently of the model.

Unit heterogeneity is also known in laboratory data: 6–19% of tests could not be mapped accurately to LOINC in oncology data [14], harmonisation reduced 2,019 unit expressions to about 40 reference units [15], and automated LOINC mapping [16] and unit harmonisation [17] have been proposed. These works target data reuse; here, an unconverted unit caused a safety failure inside the rule layer.

The contribution is not a new language model but a safety architecture for rule-consistent patient-facing explanation, and it addresses three gaps. First, studies of language models on medical results evaluate answer quality without isolating false reassurance, although that error is the one that removes the reason to seek care. Second, controlled decoding enforces the structure of an output but not its agreement with a clinical label assigned outside the model. Third, unit harmonisation is treated as an interoperability problem, whereas here an unconverted unit is a safety failure inside the rule layer itself. The paper does not propose a new language model and does not claim clinical benefit.

Rules that decide and a model that only renders those decisions is an established pattern rather than a new one. Guardrail toolkits place programmable checks around a language model [18], and the data-to-text literature has long separated content selection from surface realisation [19]. Work on data-to-document generation reports that neural generators misstate facts present in their input [20], and accuracy evaluation in that field relies on human protocols rather than automatic scores [21]. Our pipeline inherits that division of labour and adds nothing to it architecturally. What we contribute is narrower: false reassurance measured as a separate class, the cost of the pass rate reported alongside it, and a quantified failure of the deterministic layer under Vietnamese unit conventions.

3 Rule-Constrained Pipeline

The design principle is that no clinical judgement is delegated to the language model. Fig. 1 shows the four layers: Layers 1, 2, and 4 are deterministic, and only Layer 3 uses the model.

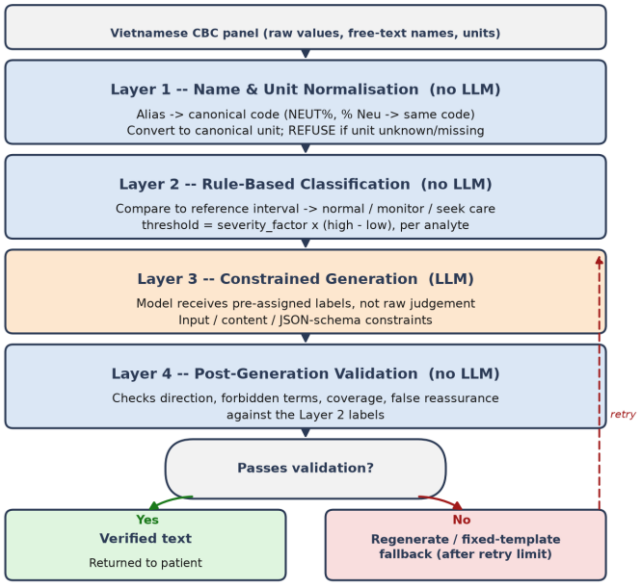


Fig. 1. Four-layer pipeline; clinical labels are fixed before generation and verified after it.

3.1 Normalisation and Rule-Based Classification

Layer 1 maps analyte aliases, such as WBC, SLBC, or the full Vietnamese name, to canonical codes; the % and # markers are kept as tokens so that NEUT% and NEUT# do not collapse into one key. Values are then converted to canonical units. Units matter: haemoglobin reported as 14 g/dL is normal, but compared directly with a 130–175 g/L interval it is classified as severe anaemia. Layer 1 therefore refuses an analyte whose unit is unrecognised or missing when units of different magnitude are possible, and reports it separately.

Layer 2 compares each value x_j with its reference interval $[l_j, h_j]$ and assigns normal, monitor, or seek care. A value outside the interval is labelled monitor if its distance d_j to the nearest bound satisfies

$$d_j \leq s_j \, (h_j - l_j), \tag{1}$$

and seek care otherwise, where sj is an analyte-specific severity factor between 0.15 and 0.50 for the 22 analytes. The factors were set heuristically, not from a critical-value list. The panel label is the most severe analyte label. Classification is therefore exact with respect to the configured intervals and severity factors, and cannot be altered by the model; it is not a claim that those thresholds are clinically validated. The names monitor and seek care denote deviation categories computed from those intervals, not triage advice: mapping a deviation category to patient-facing wording would require validation against a laboratory critical-value policy and clinician review before deployment. The reference intervals are those printed by one Vietnamese laboratory for adults, and the severity factors are ours. Both are listed per

analyte, with their provenance, in the supplementary table nguon_khoang_tham_chieu.csv.

3.2 Constrained Generation and Validation

Layer 3 runs Qwen2.5-3B locally through Ollama, so laboratory data never leaves the institution. For each analyte, the prompt supplies the value, the interval, the pre-assigned direction and label, and a plain-language description. Three constraints apply: the model receives labels instead of deriving them; it must not name diseases, causes, medications, or treatments; and it must return a fixed JSON schema.

Layer 4 compares each generated item with the Layer 2 labels and detects six error classes: false reassurance in the closing statement, direction mismatch in the structured status field, contradictory wording in the prose, forbidden terms, fabricated analytes, and omitted analytes. Before pattern matching, text is lower-cased, Vietnamese diacritics are removed, and phrases naming the reference interval, such as "higher than normal", are masked so that "normal" is not read as a status claim. A failing output is regenerated up to a limit and then replaced by a fixed template stating value, interval, and label, so no unvalidated model text is returned. The checks are lexical and therefore bound by their patterns: an output can satisfy all six and still mislead through indirect phrasing. Appendix B in the supplementary archive gives the validator algorithm, the pattern table for each error class, the masking rules, and worked pass and fail cases.

4 Experimental Design

4.1 Synthetic Data and Configurations

The benchmark contains 150 synthetic complete blood count panels with 10–18 analytes each (mean 14.1, 2,118 readings in total), divided equally into normal, monitor, and seek-care panels. Synthetic data provides known labels and a balanced design, and avoids handling patient records. Aliases and units are sampled at random to exercise Layer 1; values are rounded as laboratories print them, and panels whose rounded values leave the intended class are regenerated. Panels carry 1.5 abnormal analytes on average (range 0–4), split 124 high and 102 low across the benchmark, and 46 panels contain at least one borderline value. The generator is seeded, so the benchmark regenerates exactly; it is supplied with the analyte table, reference intervals, unit conversions, alias list, severity factors and the generated panels in the supplementary archive, and Appendix A reports the full distributions.

Seven configurations are compared on the same panels. Baseline gives raw values and intervals to the model with no labels, constraints, or validation, approximating a patient pasting a report into a chatbot. Template only uses Layers 1 and 2 and prints the fixed template directly, with no language model at all; it is the deterministic floor against which any model-based configuration has to justify itself. Constrained uses

Layers 1–3 without validation. Full uses all four layers, and Full without retry is Full with regeneration disabled, so a first failure falls straight back to the template.
The constrained configuration bundles three mechanisms: rule-assigned labels, the content prohibitions in the prompt, and the fixed JSON schema. To separate them, three further configurations were run on the same panels with the same model. Labels only supplies the labels and the schema, with the prohibitions removed. Constraints only keeps the prohibitions and the schema but gives raw values, as in the baseline. Validator only uses the baseline prompt and adds Layer 4: validation, regeneration up to the retry limit, and template fallback. Template only and Full without retry answer the two remaining questions: what the language model is worth against the rules alone, and what the regeneration loop is worth against a single attempt.

4.2 Metrics

The validator-pass rate is the proportion of panels whose final output passes the implemented Layer 4 checks, including fallback templates; model-generated pass counts only panels passed by model text. The name is deliberate. The validator is both the filter and the measure, so a high rate shows that nothing was returned that violates the implemented checks, not that the output is clinically appropriate or comprehensible, and it is not called safe output for that reason. Error metrics count panels with at least one instance of each of the six classes. Formatting errors, namely display names instead of canonical codes and unparseable JSON, are reported separately. Runtime was measured on an Intel Core i5-9400F with 24 GB RAM and no GPU. Generation used Ollama with qwen2.5:3b at Q4_K_M quantisation, temperature 0.2, a 2,048-token limit, JSON output mode, and no fixed seed. Each configuration was run once on the 150 panels. Proportions are reported with 95% Wilson confidence intervals, and configurations are compared with the McNemar test, since every configuration sees the same panels.

5 Experimental Results

Table 2 reports the comparison. Every error metric decreases monotonically from baseline to constrained to full.

Table 2. Comparison of the three main configurations on 150 synthetic panels (%).

Metric	Baseline	Constrained	Full
Safe output	0.0	48.0	100.0
Model-generated pass	0.0	48.0	64.0
False reassurance	32.0	8.0	0.0
Direction mismatch	91.3	12.7	0.0
Contradictory wording	56.0	5.3	0.0
Forbidden terms	79.3	15.3	0.0
Fabricated analytes	32.0	10.0	0.0
Omitted analytes	71.3	28.7	0.0

Mean time per panel (s)	38.9	35.8	68.3

Unconstrained generation was unsafe. Baseline passed the validator on no panel, misstated at least one direction in 91.3% of panels (95% CI 85.7–94.9) and asserted that all results were normal on 32.0% (25.1–39.8), roughly one patient in three. Pre-assigned labels and prompt constraints produced the largest drop, reducing direction mismatch to 12.7% (8.3–18.9), forbidden terms to 15.3% (10.4–22.0), contradictory wording to 5.3%, and false reassurance to 8.0% (4.6–13.5); the change in pass rate from baseline is significant (McNemar, $p < 0.001$). Nevertheless, 52.0% of constrained outputs still failed validation.

Full passed the validator on all 150 panels (95% CI 97.5–100.0). The figure needs qualification twice over. First, 64.0% of panels (56.1–71.2) were served by model text and 36.0% by fixed templates after regeneration failed, so the rate is a property of the pipeline and not evidence that a 3B model writes reliably; regeneration raised the model-written share from 46.0% (38.2–54.0) without retry to 64.0% with it, at roughly double the latency. Second, a configuration with no model at all reaches the same rate, which is taken up below.

In baseline, 97.3% of outputs used display names instead of canonical codes and 2.0% were unparseable; both fell to zero under the schema constraint. Full required 1.97 generation attempts on average and 68.3 s per panel (SD 33.7; p50 66.4, p95 121.5, p99 134.1), about twice the baseline latency of 38.9 s (p95 63.6). Re-scoring the outputs that Layer 4 rejected shows what regeneration was called on to fix: of the 54 panels that ended in a template, 66.7% had omitted an analyte, 38.9% contained a forbidden term, 29.6% misstated a direction and 18.5% falsely reassured.

One configuration detail deserves scrutiny. In the baseline, 97.3% of outputs identified analytes by display name instead of the canonical code, which raises the question of how much of its 0.0% safe-output rate reflects formatting rather than content. Every baseline output was therefore re-scored after mapping each returned name back to its canonical code through the alias table, which removes code-format errors entirely. Safe output rises from 0.0% to 0.7%, that is from zero to one panel in 150. Direction mismatch rises from 91.3% to 92.7%, because items previously counted as fabricated are now resolved to a real analyte and judged on their stated direction. False reassurance (32.0%) and contradictory wording (56.0%) are unchanged. The baseline fails on content, not on formatting.

Section 3.1 treats unit heterogeneity as a safety problem rather than a data-reuse problem, and the claim can be quantified on the same panels. Of 2,118 analyte readings, 143 (6.8%) were reported in a unit of different magnitude from the canonical one. Had the pipeline compared those readings against the reference interval without conversion, 135 of them (94.4%) would have received the wrong label, affecting 97 of 150 panels (64.7%). All 135 errors were false alarms at the most severe level; a normal haemoglobin of 14.56 g/dL, for example, is classified as seek care against a 130–175 g/L interval. None ran in the safer direction. The failure is confined to the deterministic layer and is invisible to Layer 4, because every later stage faithfully reproduces the label.

Table 3 separates the mechanisms. Labels carry the correctness improvement: supplying them alone cuts direction mismatch from 91.3% to 14.0% (9.3–20.5) and yields a 38.7% pass rate (31.2–46.7) where the baseline yields none. Prompt constraints alone do not: direction mismatch stays at 87.3% and false reassurance rises to 43.3% (35.7–51.3), above the baseline, while their one measurable effect is on vocabulary, with forbidden terms falling from 79.3% to 35.3%. Instructing the model to be careful makes it speak more cautiously, not more correctly. Adding the prohibitions on top of the labels raises the pass rate from 38.7% to 48.0%, a difference that does not reach significance on 150 panels (McNemar, $p = 0.082$). The validator alone reaches 100.0%, but every panel falls back to a template: 0.0% of panels were served by model text, against 64.0% in the full pipeline. Without labels to check against, Layer 4 can still enforce the checks, but only by discarding the explanation and returning what amounts to the original report.

Table 3. Contribution of each mechanism, 150 synthetic panels, values as % of panels except the last column. Template only uses no language model.

Configuration	Validator pass	Model text	Direction	False reass.	Forbidden	Time (s)
Baseline	0.0	0.0	91.3	32.0	79.3	38.9
Template only	100.0	0.0	0.0	0.0	0.0	0.003
Labels only	38.7	38.7	14.0	10.7	46.0	38.4
Constraints only	0.0	0.0	87.3	43.3	35.3	27.0
Validator only	100.0	0.0	0.0	0.0	0.0	95.5
Constrained	48.0	48.0	12.7	8.0	15.3	35.8
Full, no retry	100.0	46.0	0.0	0.0	0.0	35.2
Full	100.0	64.0	0.0	0.0	0.0	68.3

One comparison bounds what the language model contributes. The template-only configuration, which prints the Layer 2 labels directly and never calls a model, passes the validator on all 150 panels in 0.38 s for the whole benchmark, against 68.3 s per panel for the full pipeline. The McNemar test on the paired panels cannot distinguish the two ($p = 1.000$), and the same holds for the validator-only configuration. This is not a defect of the implementation but of the measure: a pass rate defined by consistency with the rules is maximised by output that the rules wrote. What the full pipeline adds over the template is the 64.0% of panels that keep generated prose rather than a fixed sentence, and whether that prose is easier for a patient to read is precisely the question this evaluation does not answer. The row is reported so that the comparison is visible rather than implied.

The error classes are easier to judge from instances than from rates. In panel SYN-0087, which the rules labelled monitor, the baseline closing statement read “Tất cả các chỉ số đều nằm trong khoảng bình thường” (all indices are within the normal range) and then named two abnormal ones; it is counted as false reassurance, and it shows why the closing statement is checked apart from the per-analyte fields. In panel SYN-0149, labelled seek care, the same configuration described platelets as within range when the rules had labelled them low, omitted red blood cells altogether, and

wrote “dấu hiệu của” (a sign of) about basophils, which the forbidden-term list blocks as diagnostic phrasing. Fabrication appears as well: on a normal panel the model reported a neutrophil count of 8.78 $\times 10^9$/L as above normal, although neutrophils were not among the analytes supplied. One worked example per class, with the input excerpt, the rule label, the generated text, the validator decision and the replacement output, is listed in the supplementary archive.

6 Discussion and Application Prospects

6.1 Implications for Health Informatics

The results support a division of labour rather than better prompting: comparison against a reference interval is left to rules, and paraphrasing to the model. The two components can then improve independently; replacing heuristic thresholds with a laboratory critical-value list changes only the labels, and a larger model would raise the model-written share without changing what Layer 4 enforces.

False reassurance deserves separate reporting. It may score well on readability because it is simple and calming, yet it was neither rare nor removed by instruction and reached zero only with mechanical verification. A prompt-level prohibition should not be taken as evidence of its absence.

Unit heterogeneity is a safety problem, not only a data-reuse problem. The g/dL-versus-g/L failure occurred in the deterministic layer with complete internal consistency, and no downstream check could detect it; standards such as LOINC and HL7 FHIR are therefore preconditions for rule-based clinical logic in Vietnamese laboratory reporting. A CPU runtime of 68.3 s per panel, with a 95th percentile of 121.5 s, is acceptable for a once-per-report task and avoids GPU cost for provincial hospitals; for the 36.0% fallback panels, output is no better than the report, although it never conflicts with the assigned labels.

6.2 Study Limitations

No medical professional reviewed the outputs, so they are shown to be consistent with the assigned labels, not clinically appropriate or comprehensible. The validator is both the filter and the measure, so its pass rate is not independent evidence of safety: a deterministic template reaches the same rate, and an output can satisfy every implemented check and still mislead through indirect phrasing or missing context. The severity factors are heuristic and are not clinical recommendations, and monitor and seek care are deviation categories rather than triage advice. Panels are synthetic and lack the noise, missing fields, and format variation of real reports, and Layer 4 matches surface patterns that indirect phrasing can evade.

Only structured complete blood count panels are supported; one-sided intervals and qualitative results are not handled, and results come from one 3B model. Each configuration was run once, so the intervals above carry the sampling uncertainty of 150 panels but not the decoding variability of repeated runs. Sex- and age-specific

intervals are not handled either, which matters most for haemoglobin, haematocrit and red cell count: a woman result may be labelled abnormal against a male-derived interval. Independent expert review, a patient comprehension study, validated critical-value thresholds, repeated decoding runs, larger models, broader test panels, and evaluation on real reports from multiple laboratories remain future work.

7 Conclusion

This study presents a pipeline in which rules fix every clinical label before a language model paraphrases it and a validator checks the result. On 150 synthetic panels, false reassurance fell from 32.0% under unconstrained generation to 8.0% with prompt constraints and to zero under the implemented checks, with 64.0% of panels served by model text and 36.0% by templates. A deterministic template reaches the same pass rate with no model, so the result should be read as rule consistency under the implemented checks and not as clinical safety. The contribution is an auditable architecture and a separately reported false-reassurance metric; independent expert review and a patient comprehension study are required before any patient-facing use.

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